Score vs gaze/PSM features / position
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | +0.66 | -0.05 | +0.03 | -0.02 | +0.43 | +0.55 | -0.07 | +0.51 | +0.48 |
| A: Whole task (R) | -0.43 | +0.30 | +0.30 | +0.12 | +0.35 | +0.35 | +0.15 | -0.06 | -0.06 |
| A: Reach Ring (L) | +0.43 | +0.26 | +0.13 | +0.26 | -0.16 | -0.24 | +0.32 | -0.11 | -0.22 |
| A: Reach Ring (R) | +0.04 | -0.16 | +0.24 | +0.15 | +0.12 | +0.33 | -0.03 | +0.35 | +0.11 |
| A: Grab Ring (L) | +0.27 | +0.01 | -0.15 | +0.13 | -0.21 | +0.03 | +0.18 | -0.48 | +0.10 |
| A: Grab Ring (R) | +0.52 | -0.27 | -0.16 | +0.17 | +0.43 | +0.20 | +0.36 | +0.02 | +0.19 |
| A: Transfer Ring (L) | +0.57 | +0.00 | -0.08 | +0.34 | +0.35 | +0.54 | +0.46 | +0.35 | +0.40 |
| A: Transfer Ring (R) | +0.21 | -0.78 | -0.79 | -0.02 | +0.47 | +0.38 | -0.01 | +0.16 | +0.07 |
| A: Place Ring (L) | +0.01 | +0.01 | -0.32 | +0.32 | -0.03 | +0.02 | +0.25 | +0.15 | -0.19 |
| A: Place Ring (R) | +0.08 | +0.18 | +0.25 | -0.36 | +0.03 | +0.44 | -0.33 | -0.21 | +0.32 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | -0.30 | -0.24 | +0.29 | +0.26 | +0.05 | -0.16 | +0.17 | -0.03 | -0.20 |
| C: Whole task (R) | +0.62 | +0.06 | -0.44 | -0.00 | -0.23 | +0.15 | -0.13 | -0.23 | +0.23 |
| C: Reach String (L) | -0.50 | -0.08 | +0.02 | -0.18 | +0.21 | -0.11 | -0.10 | +0.13 | -0.08 |
| C: Reach String (R) | +0.22 | -0.12 | +0.00 | -0.08 | -0.14 | -0.19 | -0.03 | -0.14 | -0.23 |
| C: Threading (L) | -0.41 | -0.20 | +0.61 | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | -0.58 | +0.08 | -0.41 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | +0.44 | -0.40 | -0.73 | -0.01 | +0.06 | -0.03 | -0.25 | -0.08 | -0.04 |
| C: Wrap around PSM (R) | +0.56 | +0.30 | -0.56 | +0.17 | +0.17 | +0.16 | -0.08 | +0.06 | -0.13 |
| C: Pull/cinch (L) | +0.29 | -0.36 | +0.71 | low N | low N | low N | low N | low N | low N |
| C: Pull/cinch (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

Time vs gaze/PSM features / position
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | +0.62 | -0.11 | -0.07 | +0.01 | +0.41 | +0.47 | -0.02 | +0.45 | +0.35 |
| A: Whole task (R) | -0.70 | +0.00 | +0.00 | +0.09 | +0.38 | +0.38 | +0.10 | +0.02 | +0.02 |
| A: Reach Ring (L) | +0.60 | +0.25 | -0.05 | +0.24 | +0.04 | -0.40 | +0.27 | +0.15 | -0.43 |
| A: Reach Ring (R) | -0.42 | -0.28 | +0.30 | +0.12 | +0.25 | +0.40 | -0.07 | +0.54 | +0.24 |
| A: Grab Ring (L) | +0.14 | -0.05 | -0.27 | +0.10 | -0.03 | -0.12 | +0.13 | -0.18 | -0.13 |
| A: Grab Ring (R) | +0.24 | -0.41 | +0.08 | +0.28 | +0.57 | +0.23 | +0.48 | +0.31 | +0.23 |
| A: Transfer Ring (L) | +0.55 | +0.08 | -0.36 | +0.38 | +0.19 | +0.39 | +0.50 * | +0.09 | +0.15 |
| A: Transfer Ring (R) | -0.29 | -0.72 | -0.74 | -0.07 | +0.43 | +0.35 | -0.09 | +0.09 | +0.02 |
| A: Place Ring (L) | +0.15 | +0.05 | -0.39 | +0.26 | -0.05 | -0.04 | +0.16 | +0.11 | -0.27 |
| A: Place Ring (R) | -0.22 | +0.19 | +0.09 | -0.47 | -0.06 | +0.36 | -0.49 | -0.33 | +0.20 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | -0.08 | -0.46 | +0.15 | +0.12 | +0.37 | +0.13 | -0.01 | +0.31 | +0.25 |
| C: Whole task (R) | +0.27 | -0.06 | +0.56 | +0.51 * | -0.28 | +0.11 | +0.55 * | -0.30 | -0.06 |
| C: Reach String (L) | -0.21 | -0.10 | +0.05 | +0.46 | +0.13 | +0.03 | +0.65 * | +0.16 | +0.02 |
| C: Reach String (R) | -0.06 | -0.32 | -0.01 | +0.23 | -0.04 | +0.51 * | +0.25 | -0.12 | +0.50 |
| C: Threading (L) | low N | low N | low N | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | -0.49 | +0.31 | -0.60 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | +0.39 | -0.14 | -0.25 | +0.37 | +0.45 | +0.07 | +0.37 | +0.43 | +0.11 |
| C: Wrap around PSM (R) | +0.94 * | +0.43 | -0.24 | +0.47 | +0.00 | +0.28 | +0.38 | -0.14 | +0.10 |
| C: Pull/cinch (L) | +0.09 | -0.15 | +0.32 | +0.49 | +0.06 | -0.08 | +0.49 | +0.21 | -0.11 |
| C: Pull/cinch (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

Eco vs gaze/PSM features / position
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | +0.33 | +0.32 | +0.41 | -0.17 | +0.17 | +0.43 | -0.26 | +0.09 | +0.29 |
| A: Whole task (R) | -0.19 | +0.39 | +0.39 | +0.18 | +0.44 | +0.44 | +0.21 | +0.17 | +0.17 |
| A: Reach Ring (L) | +0.27 | +0.06 | +0.41 | +0.21 | -0.00 | -0.06 | +0.22 | +0.07 | +0.01 |
| A: Reach Ring (R) | +0.31 | -0.23 | +0.03 | +0.10 | +0.27 | +0.44 | -0.08 | +0.50 | +0.30 |
| A: Grab Ring (L) | +0.35 | -0.12 | -0.16 | +0.24 | -0.46 | -0.10 | +0.28 | -0.62 | -0.09 |
| A: Grab Ring (R) | +0.42 | +0.11 | -0.07 | +0.27 | +0.35 | +0.15 | +0.43 | -0.01 | +0.12 |
| A: Transfer Ring (L) | +0.42 | +0.21 | +0.07 | +0.29 | +0.16 | +0.36 | +0.34 | +0.07 | +0.12 |
| A: Transfer Ring (R) | +0.55 | -0.59 | -0.59 | +0.09 | +0.37 | +0.36 | +0.14 | +0.04 | +0.09 |
| A: Place Ring (L) | -0.19 | +0.02 | -0.09 | +0.15 | +0.04 | +0.22 | +0.02 | +0.20 | +0.14 |
| A: Place Ring (R) | +0.26 | +0.28 | +0.09 | -0.31 | -0.11 | +0.25 | -0.24 | -0.34 | +0.05 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | +0.09 | -0.46 | +0.13 | +0.00 | +0.41 | +0.26 | -0.13 | +0.35 | +0.39 |
| C: Whole task (R) | -0.10 | -0.07 | -0.03 | +0.39 | -0.18 | +0.12 | +0.41 | -0.17 | -0.03 |
| C: Reach String (L) | -0.45 | -0.01 | -0.14 | +0.45 | +0.05 | +0.12 | +0.64 | +0.06 | +0.13 |
| C: Reach String (R) | -0.17 | -0.14 | -0.43 | +0.14 | +0.05 | +0.54 | +0.15 | -0.00 | +0.53 |
| C: Threading (L) | low N | low N | low N | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | +0.14 | +0.94 | -0.09 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | +0.22 | -0.28 | -0.70 | +0.28 | +0.35 | +0.05 | +0.27 | +0.32 | +0.08 |
| C: Wrap around PSM (R) | +0.68 | +0.21 | -0.41 | +0.37 | -0.12 | +0.27 | +0.27 | -0.28 | +0.08 |
| C: Pull/cinch (L) | -0.18 | -0.16 | +0.27 | -0.01 | -0.30 | +0.21 | -0.06 | +0.16 | +0.14 |
| C: Pull/cinch (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

TimeRaw vs gaze/PSM features / position
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | -0.62 | +0.11 | +0.07 | -0.00 | -0.41 | -0.48 | +0.03 | -0.46 | -0.35 |
| A: Whole task (R) | +0.70 | +0.00 | +0.00 | -0.09 | -0.38 | -0.38 | -0.09 | -0.02 | -0.02 |
| A: Reach Ring (L) | -0.60 | -0.25 | +0.05 | -0.25 | -0.03 | +0.40 | -0.29 | -0.13 | +0.43 |
| A: Reach Ring (R) | +0.42 | +0.28 | -0.30 | -0.11 | -0.25 | -0.40 | +0.09 | -0.53 | -0.24 |
| A: Grab Ring (L) | -0.14 | +0.05 | +0.27 | -0.09 | +0.04 | +0.12 | -0.13 | +0.21 | +0.13 |
| A: Grab Ring (R) | -0.24 | +0.41 | -0.08 | -0.29 | -0.57 | -0.22 | -0.48 | -0.30 | -0.21 |
| A: Transfer Ring (L) | -0.55 | -0.08 | +0.36 | -0.37 | -0.19 | -0.39 | -0.49 * | -0.10 | -0.15 |
| A: Transfer Ring (R) | +0.29 | +0.72 | +0.74 | +0.06 | -0.42 | -0.33 | +0.08 | -0.08 | -0.00 |
| A: Place Ring (L) | -0.15 | -0.05 | +0.39 | -0.26 | +0.04 | +0.04 | -0.16 | -0.13 | +0.27 |
| A: Place Ring (R) | +0.22 | -0.19 | -0.09 | +0.47 | +0.07 | -0.36 | +0.49 | +0.33 | -0.20 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | +0.08 | +0.46 | -0.15 | -0.12 | -0.37 | -0.13 | +0.01 | -0.31 | -0.25 |
| C: Whole task (R) | -0.27 | +0.06 | -0.56 | -0.51 * | +0.28 | -0.11 | -0.55 * | +0.30 | +0.06 |
| C: Reach String (L) | +0.21 | +0.10 | -0.05 | -0.46 | -0.13 | -0.03 | -0.65 * | -0.16 | -0.02 |
| C: Reach String (R) | +0.06 | +0.32 | +0.01 | -0.23 | +0.04 | -0.51 * | -0.25 | +0.12 | -0.50 |
| C: Threading (L) | low N | low N | low N | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | +0.49 | -0.31 | +0.60 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | -0.39 | +0.14 | +0.25 | -0.37 | -0.45 | -0.07 | -0.37 | -0.43 | -0.11 |
| C: Wrap around PSM (R) | -0.94 * | -0.43 | +0.24 | -0.47 | -0.00 | -0.28 | -0.38 | +0.14 | -0.10 |
| C: Pull/cinch (L) | -0.09 | +0.15 | -0.32 | -0.49 | -0.06 | +0.08 | -0.49 | -0.21 | +0.11 |
| C: Pull/cinch (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

EcoRaw vs gaze/PSM features / position
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | -0.33 | -0.32 | -0.41 | +0.17 | -0.16 | -0.43 | +0.26 | -0.08 | -0.28 |
| A: Whole task (R) | +0.19 | -0.39 | -0.39 | -0.17 | -0.45 | -0.45 | -0.21 | -0.17 | -0.17 |
| A: Reach Ring (L) | -0.27 | -0.06 | -0.41 | -0.20 | +0.00 | +0.05 | -0.21 | -0.07 | -0.02 |
| A: Reach Ring (R) | -0.31 | +0.23 | -0.03 | -0.10 | -0.27 | -0.44 | +0.08 | -0.51 | -0.30 |
| A: Grab Ring (L) | -0.35 | +0.12 | +0.16 | -0.24 | +0.46 | +0.10 | -0.28 | +0.62 | +0.09 |
| A: Grab Ring (R) | -0.42 | -0.11 | +0.07 | -0.27 | -0.36 | -0.16 | -0.42 | +0.00 | -0.13 |
| A: Transfer Ring (L) | -0.42 | -0.21 | -0.07 | -0.29 | -0.15 | -0.36 | -0.34 | -0.05 | -0.12 |
| A: Transfer Ring (R) | -0.55 | +0.59 | +0.59 | -0.09 | -0.37 | -0.37 | -0.14 | -0.05 | -0.09 |
| A: Place Ring (L) | +0.19 | -0.02 | +0.09 | -0.15 | -0.04 | -0.23 | -0.02 | -0.20 | -0.15 |
| A: Place Ring (R) | -0.26 | -0.28 | -0.09 | +0.30 | +0.11 | -0.24 | +0.23 | +0.34 | -0.04 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | -0.09 | +0.46 | -0.13 | -0.00 | -0.41 | -0.26 | +0.13 | -0.35 | -0.39 |
| C: Whole task (R) | +0.10 | +0.07 | +0.03 | -0.39 | +0.18 | -0.12 | -0.41 | +0.17 | +0.03 |
| C: Reach String (L) | +0.45 | +0.01 | +0.14 | -0.45 | -0.05 | -0.12 | -0.64 | -0.06 | -0.13 |
| C: Reach String (R) | +0.17 | +0.14 | +0.43 | -0.14 | -0.05 | -0.54 | -0.15 | +0.00 | -0.53 |
| C: Threading (L) | low N | low N | low N | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | -0.14 | -0.94 | +0.09 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | -0.22 | +0.28 | +0.70 | -0.28 | -0.35 | -0.05 | -0.27 | -0.32 | -0.08 |
| C: Wrap around PSM (R) | -0.68 | -0.21 | +0.41 | -0.37 | +0.12 | -0.27 | -0.27 | +0.28 | -0.08 |
| C: Pull/cinch (L) | +0.18 | +0.16 | -0.27 | +0.01 | +0.30 | -0.21 | +0.06 | -0.16 | -0.14 |
| C: Pull/cinch (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

PenaltySum vs gaze/PSM features / position
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | +0.38 | +0.21 | +0.41 | -0.04 | +0.37 | +0.49 | -0.08 | +0.37 | +0.37 |
| A: Whole task (R) | -0.28 | +0.37 | +0.37 | +0.11 | +0.33 | +0.33 | +0.10 | +0.04 | +0.04 |
| A: Reach Ring (L) | +0.21 | +0.05 | +0.38 | +0.34 | -0.30 | +0.04 | +0.36 | -0.28 | +0.12 |
| A: Reach Ring (R) | +0.17 | -0.28 | +0.30 | +0.06 | -0.17 | +0.17 | -0.09 | -0.14 | -0.06 |
| A: Grab Ring (L) | +0.21 | -0.01 | -0.29 | -0.12 | +0.17 | +0.38 | -0.15 | +0.13 | +0.47 |
| A: Grab Ring (R) | +0.55 | -0.24 | -0.25 | -0.12 | +0.21 | -0.09 | -0.08 | -0.31 | -0.22 |
| A: Transfer Ring (L) | +0.42 | -0.06 | -0.08 | +0.18 | +0.33 | +0.43 | +0.17 | +0.29 | +0.25 |
| A: Transfer Ring (R) | +0.28 | -0.66 | -0.66 | +0.05 | +0.10 | +0.03 | +0.08 | -0.31 | -0.36 |
| A: Place Ring (L) | +0.03 | -0.12 | -0.51 | +0.29 | -0.12 | +0.15 | +0.21 | -0.02 | +0.04 |
| A: Place Ring (R) | +0.28 | +0.26 | +0.31 | +0.09 | +0.16 | +0.32 | +0.26 | +0.05 | +0.17 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | -0.02 | -0.65 | +0.09 | +0.24 | +0.37 | +0.18 | +0.14 | +0.29 | +0.34 |
| C: Whole task (R) | +0.33 | -0.31 | -0.11 | -0.14 | -0.12 | +0.10 | -0.09 | -0.01 | +0.10 |
| C: Reach String (L) | -0.41 | -0.29 | -0.36 | -0.32 | +0.27 | -0.20 | -0.15 | +0.22 | -0.20 |
| C: Reach String (R) | -0.08 | -0.03 | -0.24 | -0.11 | +0.09 | -0.22 | -0.10 | +0.05 | -0.38 |
| C: Threading (L) | low N | low N | low N | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | -0.71 | -0.20 | -0.31 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | +0.41 | +0.01 | -0.62 | +0.31 | +0.05 | +0.13 | +0.22 | -0.06 | +0.30 |
| C: Wrap around PSM (R) | +0.62 | +0.25 | -0.07 | +0.46 | +0.48 | +0.22 | +0.27 | +0.37 | -0.10 |
| C: Pull/cinch (L) | +0.31 | -0.39 | +0.87 | +0.10 | -0.72 | +0.07 | +0.11 | -0.12 | -0.56 |
| C: Pull/cinch (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

PenaltyRawSum vs gaze/PSM features / position
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | -0.51 | +0.13 | -0.05 | -0.04 | -0.22 | -0.33 | -0.03 | -0.17 | -0.18 |
| A: Whole task (R) | +0.35 | -0.15 | -0.15 | -0.07 | -0.25 | -0.25 | -0.06 | +0.02 | +0.02 |
| A: Reach Ring (L) | -0.25 | -0.09 | -0.24 | -0.31 | +0.37 | -0.08 | -0.31 | +0.35 | -0.12 |
| A: Reach Ring (R) | -0.07 | +0.27 | -0.34 | +0.03 | -0.01 | -0.11 | +0.18 | -0.08 | +0.10 |
| A: Grab Ring (L) | -0.19 | -0.11 | +0.36 | +0.25 | +0.07 | -0.40 | +0.29 | +0.15 | -0.48 |
| A: Grab Ring (R) | -0.56 | +0.36 | +0.28 | +0.15 | -0.17 | -0.01 | +0.12 | +0.21 | +0.05 |
| A: Transfer Ring (L) | -0.55 | +0.22 | +0.27 | -0.09 | -0.29 | -0.43 | -0.06 | -0.23 | -0.27 |
| A: Transfer Ring (R) | -0.19 | +0.80 | +0.81 | -0.08 | -0.03 | +0.00 | -0.11 | +0.31 | +0.31 |
| A: Place Ring (L) | -0.27 | +0.13 | +0.59 | -0.16 | -0.13 | -0.24 | -0.04 | -0.29 | -0.17 |
| A: Place Ring (R) | -0.13 | -0.10 | -0.51 | -0.13 | -0.10 | -0.20 | -0.29 | +0.01 | -0.04 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | +0.06 | +0.72 | +0.10 | -0.21 | -0.20 | +0.07 | -0.13 | -0.13 | +0.00 |
| C: Whole task (R) | -0.11 | +0.24 | -0.03 | -0.17 | +0.18 | -0.19 | -0.18 | +0.17 | -0.08 |
| C: Reach String (L) | +0.17 | +0.37 | +0.19 | -0.52 | -0.04 | +0.22 | -0.70 | -0.04 | +0.24 |
| C: Reach String (R) | +0.08 | +0.33 | +0.25 | -0.30 | -0.01 | -0.27 | -0.33 | +0.08 | -0.25 |
| C: Threading (L) | low N | low N | low N | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | +0.43 | -0.43 | +0.66 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | -0.27 | +0.04 | +0.64 | -0.42 | -0.47 | -0.15 | -0.43 | -0.46 | -0.18 |
| C: Wrap around PSM (R) | -0.85 | -0.48 | +0.28 | -0.51 | -0.07 | -0.42 | -0.45 | +0.03 | -0.33 |
| C: Pull/cinch (L) | +0.24 | +0.28 | -0.44 | -0.38 | +0.53 | +0.26 | -0.43 | +0.01 | +0.63 |
| C: Pull/cinch (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

Score vs gaze/PSM features / velocity
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | +0.78 | +0.34 | -0.02 | +0.34 | +0.25 | +0.10 | +0.21 | +0.25 | +0.20 |
| A: Whole task (R) | +0.64 | +0.13 | -0.42 | +0.67 | +0.01 | -0.02 | +0.24 | +0.19 | +0.31 |
| A: Reach Ring (L) | +0.39 | +0.47 | +0.20 | +0.26 | +0.57 | -0.01 | +0.22 | +0.52 | -0.03 |
| A: Reach Ring (R) | +0.35 | -0.41 | -0.19 | +0.59 | +0.45 | +0.05 | +0.26 | +0.38 | +0.04 |
| A: Grab Ring (L) | +0.64 | +0.15 | -0.56 | +0.38 | -0.22 | +0.18 | +0.19 | -0.06 | -0.27 |
| A: Grab Ring (R) | +0.40 | -0.35 | -0.39 | +0.45 | +0.26 | -0.16 | +0.26 | -0.09 | +0.09 |
| A: Transfer Ring (L) | +0.71 | -0.12 | -0.24 | +0.64 | -0.00 | +0.21 | +0.33 | -0.10 | -0.30 |
| A: Transfer Ring (R) | +0.52 | -0.04 | +0.01 | +0.45 | +0.17 | +0.01 | +0.36 | +0.09 | +0.30 |
| A: Place Ring (L) | +0.48 | +0.38 | +0.15 | +0.34 | -0.02 | +0.14 | +0.51 | +0.01 | +0.30 |
| A: Place Ring (R) | +0.31 | +0.45 | -0.26 | +0.39 | +0.10 | +0.21 | +0.43 | -0.24 | +0.02 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | +0.31 | +0.57 | +0.04 | -0.11 | -0.10 | +0.27 | -0.23 | -0.17 | +0.27 |
| C: Whole task (R) | +0.12 | -0.19 | +0.36 | +0.07 | -0.24 | +0.15 | -0.12 | -0.03 | +0.13 |
| C: Reach String (L) | +0.16 | -0.27 | +0.51 | +0.07 | -0.35 | -0.04 | +0.04 | -0.29 | -0.17 |
| C: Reach String (R) | +0.03 | -0.61 | -0.10 | +0.21 | +0.01 | -0.15 | +0.21 | +0.16 | -0.04 |
| C: Threading (L) | +0.51 | +0.76 | +0.37 | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | +0.19 | +0.22 | +0.44 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | +0.13 | +0.28 | -0.28 | +0.09 | +0.08 | +0.23 | -0.07 | +0.10 | +0.17 |
| C: Wrap around PSM (R) | +0.54 | +0.24 | -0.05 | -0.08 | -0.13 | -0.03 | -0.26 | -0.26 | -0.07 |
| C: Pull/cinch (L) | +0.21 | +0.02 | +0.17 | -0.35 | -0.00 | -0.17 | -0.11 | -0.20 | +0.12 |
| C: Pull/cinch (R) | +0.26 | +0.14 | -0.08 | +0.02 | -0.21 | +0.42 | -0.19 | -0.14 | +0.36 |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

Time vs gaze/PSM features / velocity
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | +0.63 | +0.38 | +0.14 | +0.30 | +0.24 | +0.10 | +0.14 | +0.21 | +0.19 |
| A: Whole task (R) | +0.40 | +0.33 | -0.10 | +0.59 | -0.00 | -0.16 | +0.08 | +0.16 | +0.07 |
| A: Reach Ring (L) | +0.44 | +0.21 | +0.23 | +0.30 | +0.55 | +0.08 | +0.28 | +0.46 | +0.12 |
| A: Reach Ring (R) | +0.32 | -0.19 | -0.15 | +0.58 | +0.38 | +0.01 | +0.25 | +0.26 | -0.02 |
| A: Grab Ring (L) | +0.64 | +0.03 | -0.46 | +0.42 | -0.15 | +0.26 | +0.27 | +0.06 | -0.09 |
| A: Grab Ring (R) | +0.28 | -0.52 | -0.02 | +0.41 | +0.25 | -0.09 | +0.20 | -0.08 | +0.19 |
| A: Transfer Ring (L) | +0.47 | +0.25 | -0.14 | +0.62 | -0.08 | +0.26 | +0.30 | -0.22 | -0.17 |
| A: Transfer Ring (R) | +0.31 | +0.07 | -0.16 | +0.37 | +0.14 | +0.01 | +0.23 | +0.04 | +0.27 |
| A: Place Ring (L) | +0.44 | +0.26 | +0.25 | +0.24 | +0.01 | +0.19 | +0.31 | +0.05 | +0.36 |
| A: Place Ring (R) | -0.21 | +0.29 | +0.17 | +0.38 | +0.12 | +0.21 | +0.39 | -0.19 | +0.03 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | +0.05 | -0.47 | +0.80 | +0.47 | +0.43 | -0.32 | +0.52 | +0.40 | -0.39 |
| C: Whole task (R) | +0.24 | +0.13 | +0.37 | +0.51 | -0.07 | -0.20 | +0.52 | -0.13 | -0.19 |
| C: Reach String (L) | +0.10 | +0.15 | -0.16 | +0.51 | -0.20 | +0.13 | +0.51 | -0.20 | +0.20 |
| C: Reach String (R) | +0.31 | +0.03 | +0.16 | +0.41 | -0.25 | +0.00 | +0.38 | -0.35 | +0.07 |
| C: Threading (L) | -0.03 | +0.26 | -0.31 | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | -0.26 | +0.37 | +0.66 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | +0.59 | -0.28 | +0.21 | +0.37 | -0.12 | -0.01 | +0.29 | -0.15 | -0.10 |
| C: Wrap around PSM (R) | -0.09 | -0.16 | -0.35 | +0.30 | -0.05 | +0.19 | +0.37 | -0.06 | +0.13 |
| C: Pull/cinch (L) | +0.27 | -0.14 | +0.43 | +0.10 | -0.02 | -0.03 | +0.17 | -0.03 | +0.07 |
| C: Pull/cinch (R) | +0.25 | +0.28 | +0.05 | +0.44 | -0.09 | -0.05 | +0.44 | -0.11 | -0.14 |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

Eco vs gaze/PSM features / velocity
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | +0.62 | +0.42 | -0.16 | +0.29 | +0.28 | -0.06 | +0.15 | +0.26 | -0.07 |
| A: Whole task (R) | +0.75 | -0.15 | -0.55 | +0.64 | -0.01 | -0.10 | +0.29 | +0.11 | +0.13 |
| A: Reach Ring (L) | +0.35 | +0.44 | +0.11 | +0.25 | +0.49 | +0.05 | +0.19 | +0.37 | +0.06 |
| A: Reach Ring (R) | +0.40 | -0.28 | -0.04 | +0.50 | +0.44 | -0.02 | +0.16 | +0.35 | -0.06 |
| A: Grab Ring (L) | +0.52 | +0.11 | -0.30 | +0.45 | -0.30 | +0.22 | +0.30 | -0.20 | -0.10 |
| A: Grab Ring (R) | +0.09 | -0.13 | -0.26 | +0.59 | +0.18 | -0.14 | +0.49 | -0.14 | +0.08 |
| A: Transfer Ring (L) | +0.74 | -0.13 | -0.36 | +0.61 | +0.05 | +0.07 | +0.32 | +0.01 | -0.41 |
| A: Transfer Ring (R) | +0.69 | -0.19 | -0.05 | +0.40 | +0.15 | -0.05 | +0.27 | +0.06 | +0.14 |
| A: Place Ring (L) | +0.31 | +0.27 | +0.20 | +0.14 | -0.04 | +0.25 | +0.14 | -0.03 | +0.40 |
| A: Place Ring (R) | +0.33 | +0.24 | -0.17 | +0.40 | +0.04 | +0.23 | +0.39 | -0.26 | +0.08 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | +0.17 | -0.30 | +0.32 | +0.49 | +0.41 | -0.33 | +0.54 | +0.38 | -0.40 |
| C: Whole task (R) | -0.01 | +0.35 | +0.29 | +0.56 | -0.16 | -0.13 | +0.56 | -0.22 | -0.11 |
| C: Reach String (L) | -0.06 | +0.23 | -0.15 | +0.48 | -0.17 | +0.10 | +0.49 | -0.17 | +0.16 |
| C: Reach String (R) | +0.02 | -0.46 | -0.19 | +0.50 | -0.33 | +0.08 | +0.48 | -0.43 | +0.14 |
| C: Threading (L) | +0.14 | +0.31 | -0.26 | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | -0.66 | -0.03 | +0.49 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | +0.35 | -0.44 | -0.11 | +0.37 | -0.15 | +0.04 | +0.30 | -0.18 | -0.03 |
| C: Wrap around PSM (R) | -0.09 | +0.19 | -0.07 | +0.28 | +0.02 | +0.18 | +0.35 | +0.00 | +0.12 |
| C: Pull/cinch (L) | +0.35 | -0.29 | +0.22 | +0.14 | -0.09 | +0.01 | +0.23 | -0.13 | +0.12 |
| C: Pull/cinch (R) | +0.40 | +0.07 | -0.06 | +0.46 | +0.07 | -0.08 | +0.45 | +0.05 | -0.16 |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

TimeRaw vs gaze/PSM features / velocity
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | -0.63 | -0.38 | -0.14 | -0.29 | -0.23 | -0.10 | -0.13 | -0.21 | -0.18 |
| A: Whole task (R) | -0.40 | -0.33 | +0.10 | -0.59 | +0.00 | +0.16 | -0.09 | -0.15 | -0.08 |
| A: Reach Ring (L) | -0.44 | -0.21 | -0.23 | -0.29 | -0.54 | -0.08 | -0.27 | -0.45 | -0.11 |
| A: Reach Ring (R) | -0.32 | +0.19 | +0.15 | -0.58 | -0.38 | -0.01 | -0.25 | -0.26 | +0.02 |
| A: Grab Ring (L) | -0.64 | -0.03 | +0.46 | -0.42 | +0.15 | -0.26 | -0.26 | -0.06 | +0.08 |
| A: Grab Ring (R) | -0.28 | +0.52 | +0.02 | -0.41 | -0.25 | +0.09 | -0.20 | +0.09 | -0.19 |
| A: Transfer Ring (L) | -0.47 | -0.25 | +0.14 | -0.62 | +0.09 | -0.26 | -0.30 | +0.22 | +0.16 |
| A: Transfer Ring (R) | -0.31 | -0.07 | +0.16 | -0.36 | -0.14 | -0.01 | -0.21 | -0.05 | -0.27 |
| A: Place Ring (L) | -0.44 | -0.26 | -0.25 | -0.24 | -0.01 | -0.20 | -0.32 | -0.05 | -0.37 |
| A: Place Ring (R) | +0.21 | -0.29 | -0.17 | -0.38 | -0.12 | -0.21 | -0.40 | +0.19 | -0.03 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | -0.05 | +0.47 | -0.80 | -0.47 | -0.43 | +0.32 | -0.52 | -0.40 | +0.39 |
| C: Whole task (R) | -0.24 | -0.13 | -0.37 | -0.51 | +0.07 | +0.20 | -0.52 | +0.13 | +0.19 |
| C: Reach String (L) | -0.10 | -0.15 | +0.16 | -0.51 | +0.20 | -0.13 | -0.51 | +0.20 | -0.20 |
| C: Reach String (R) | -0.31 | -0.03 | -0.16 | -0.41 | +0.25 | -0.00 | -0.38 | +0.35 | -0.07 |
| C: Threading (L) | +0.03 | -0.26 | +0.31 | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | +0.26 | -0.37 | -0.66 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | -0.59 | +0.28 | -0.21 | -0.37 | +0.12 | +0.01 | -0.29 | +0.15 | +0.10 |
| C: Wrap around PSM (R) | +0.09 | +0.16 | +0.35 | -0.30 | +0.05 | -0.19 | -0.37 | +0.06 | -0.13 |
| C: Pull/cinch (L) | -0.27 | +0.14 | -0.43 | -0.10 | +0.02 | +0.03 | -0.17 | +0.03 | -0.07 |
| C: Pull/cinch (R) | -0.25 | -0.28 | -0.05 | -0.44 | +0.09 | +0.05 | -0.44 | +0.11 | +0.14 |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

EcoRaw vs gaze/PSM features / velocity
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | -0.62 | -0.42 | +0.16 | -0.29 | -0.28 | +0.06 | -0.15 | -0.27 | +0.07 |
| A: Whole task (R) | -0.75 | +0.15 | +0.55 | -0.64 | +0.01 | +0.10 | -0.28 | -0.11 | -0.12 |
| A: Reach Ring (L) | -0.35 | -0.44 | -0.11 | -0.25 | -0.49 | -0.05 | -0.19 | -0.37 | -0.06 |
| A: Reach Ring (R) | -0.40 | +0.28 | +0.04 | -0.50 | -0.43 | +0.03 | -0.16 | -0.33 | +0.07 |
| A: Grab Ring (L) | -0.52 | -0.11 | +0.30 | -0.45 | +0.30 | -0.22 | -0.30 | +0.19 | +0.09 |
| A: Grab Ring (R) | -0.09 | +0.13 | +0.26 | -0.59 | -0.19 | +0.13 | -0.49 | +0.14 | -0.09 |
| A: Transfer Ring (L) | -0.74 | +0.13 | +0.36 | -0.61 | -0.06 | -0.07 | -0.33 | -0.01 | +0.41 |
| A: Transfer Ring (R) | -0.69 | +0.19 | +0.05 | -0.39 | -0.14 | +0.05 | -0.26 | -0.06 | -0.14 |
| A: Place Ring (L) | -0.31 | -0.27 | -0.20 | -0.13 | +0.05 | -0.25 | -0.14 | +0.04 | -0.40 |
| A: Place Ring (R) | -0.33 | -0.24 | +0.17 | -0.40 | -0.04 | -0.23 | -0.40 | +0.26 | -0.08 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | -0.17 | +0.30 | -0.32 | -0.49 | -0.41 | +0.33 | -0.54 | -0.38 | +0.40 |
| C: Whole task (R) | +0.01 | -0.35 | -0.29 | -0.56 | +0.16 | +0.13 | -0.56 | +0.22 | +0.11 |
| C: Reach String (L) | +0.06 | -0.23 | +0.15 | -0.48 | +0.17 | -0.10 | -0.49 | +0.17 | -0.16 |
| C: Reach String (R) | -0.02 | +0.46 | +0.19 | -0.50 | +0.33 | -0.08 | -0.48 | +0.43 | -0.14 |
| C: Threading (L) | -0.14 | -0.31 | +0.26 | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | +0.66 | +0.03 | -0.49 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | -0.35 | +0.44 | +0.11 | -0.37 | +0.15 | -0.04 | -0.30 | +0.18 | +0.03 |
| C: Wrap around PSM (R) | +0.09 | -0.19 | +0.07 | -0.28 | -0.02 | -0.18 | -0.35 | -0.00 | -0.12 |
| C: Pull/cinch (L) | -0.35 | +0.29 | -0.22 | -0.14 | +0.09 | -0.01 | -0.23 | +0.13 | -0.12 |
| C: Pull/cinch (R) | -0.40 | -0.07 | +0.06 | -0.46 | -0.07 | +0.08 | -0.45 | -0.05 | +0.16 |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

PenaltySum vs gaze/PSM features / velocity
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | +0.69 | +0.55 | +0.02 | +0.22 | +0.09 | -0.12 | +0.07 | +0.00 | -0.14 |
| A: Whole task (R) | +0.59 | +0.14 | -0.38 | +0.31 | -0.04 | -0.03 | -0.19 | +0.04 | +0.16 |
| A: Reach Ring (L) | +0.42 | +0.27 | +0.27 | +0.19 | +0.44 | -0.04 | +0.11 | +0.30 | -0.06 |
| A: Reach Ring (R) | +0.34 | -0.23 | -0.10 | +0.17 | +0.18 | +0.07 | -0.24 | +0.02 | +0.06 |
| A: Grab Ring (L) | +0.47 | +0.06 | -0.21 | +0.09 | -0.14 | +0.05 | -0.17 | +0.00 | -0.26 |
| A: Grab Ring (R) | +0.09 | -0.27 | -0.33 | +0.23 | +0.14 | -0.21 | -0.02 | -0.13 | -0.06 |
| A: Transfer Ring (L) | +0.68 | -0.04 | -0.25 | +0.32 | +0.03 | -0.06 | -0.09 | -0.02 | -0.49 |
| A: Transfer Ring (R) | +0.62 | -0.35 | -0.01 | +0.10 | +0.19 | +0.14 | -0.13 | +0.13 | +0.35 |
| A: Place Ring (L) | +0.62 | +0.36 | +0.15 | +0.53 | +0.04 | -0.13 | +0.64 | +0.07 | -0.14 |
| A: Place Ring (R) | +0.25 | +0.20 | -0.19 | +0.27 | -0.04 | +0.06 | +0.20 | -0.30 | -0.12 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | +0.37 | +0.05 | +0.53 | -0.47 | -0.02 | +0.16 | -0.52 | -0.22 | +0.04 |
| C: Whole task (R) | +0.01 | -0.07 | +0.45 | -0.02 | -0.11 | +0.36 | -0.15 | -0.15 | +0.48 |
| C: Reach String (L) | -0.13 | -0.27 | -0.07 | -0.19 | -0.01 | -0.43 | -0.28 | +0.03 | -0.58 |
| C: Reach String (R) | +0.12 | -0.22 | -0.32 | +0.19 | -0.02 | -0.19 | +0.03 | -0.07 | -0.06 |
| C: Threading (L) | +0.77 | +0.03 | +0.77 | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | +0.09 | +0.66 | +0.49 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | +0.15 | -0.18 | +0.23 | +0.22 | -0.05 | +0.19 | -0.09 | -0.15 | +0.09 |
| C: Wrap around PSM (R) | +0.44 | +0.08 | +0.23 | +0.03 | -0.11 | +0.16 | +0.07 | -0.22 | -0.00 |
| C: Pull/cinch (L) | +0.35 | +0.27 | +0.11 | -0.30 | +0.24 | -0.55 | -0.12 | +0.15 | -0.38 |
| C: Pull/cinch (R) | +0.25 | +0.12 | -0.18 | -0.12 | +0.07 | +0.36 | -0.26 | -0.02 | +0.27 |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
Whole-task rows appear only where the source analysis contains them. Missing support is shown explicitly.
Scores describe complete attempts. Positive signed lag means gaze leads PSM.

PenaltyRawSum vs gaze/PSM features / velocity
* BH q < 0.05 in the original outcome/scenario/method family

| | Between participants | | | Within participant | | | Within participant adjusted for task order + attempt | | |
|---|---|---|---|---|---|---|---|---|---|
| | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag | Peak r | Signed lag | Absolute lag |
| A: Whole task (L) | -0.65 | -0.53 | -0.10 | -0.10 | -0.06 | +0.11 | +0.05 | +0.01 | +0.13 |
| A: Whole task (R) | -0.57 | -0.22 | +0.44 | -0.28 | +0.06 | -0.06 | +0.18 | -0.02 | -0.25 |
| A: Reach Ring (L) | -0.23 | -0.51 | -0.13 | -0.15 | -0.43 | +0.05 | -0.07 | -0.30 | +0.07 |
| A: Reach Ring (R) | -0.25 | +0.48 | +0.31 | -0.20 | -0.18 | -0.03 | +0.15 | -0.04 | -0.02 |
| A: Grab Ring (L) | -0.45 | -0.30 | +0.39 | +0.05 | +0.23 | -0.06 | +0.30 | +0.13 | +0.19 |
| A: Grab Ring (R) | -0.18 | +0.55 | +0.48 | -0.36 | -0.05 | +0.14 | -0.19 | +0.22 | -0.01 |
| A: Transfer Ring (L) | -0.51 | -0.14 | +0.11 | -0.30 | -0.20 | +0.06 | +0.05 | -0.20 | +0.42 |
| A: Transfer Ring (R) | -0.38 | +0.06 | +0.10 | -0.12 | -0.00 | -0.10 | +0.07 | +0.10 | -0.26 |
| A: Place Ring (L) | -0.66 | -0.49 | -0.09 | -0.46 | -0.13 | +0.17 | -0.52 | -0.18 | +0.18 |
| A: Place Ring (R) | -0.09 | -0.23 | +0.31 | -0.27 | +0.06 | -0.07 | -0.21 | +0.29 | +0.09 |
| A: Drop Ring (L) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| A: Drop Ring (R) | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded | excluded |
| C: Whole task (L) | -0.12 | +0.28 | -0.61 | -0.16 | -0.32 | +0.21 | -0.22 | -0.27 | +0.28 |
| C: Whole task (R) | +0.08 | -0.33 | -0.23 | -0.54 | -0.14 | +0.04 | -0.56 | -0.07 | +0.01 |
| C: Reach String (L) | -0.04 | -0.15 | +0.13 | -0.34 | +0.15 | +0.23 | -0.34 | +0.16 | +0.17 |
| C: Reach String (R) | -0.23 | +0.15 | -0.07 | -0.18 | +0.13 | +0.18 | -0.13 | +0.27 | +0.10 |
| C: Threading (L) | +0.14 | -0.49 | +0.49 | low N | low N | low N | low N | low N | low N |
| C: Threading (R) | +0.37 | -0.20 | -0.60 | low N | low N | low N | low N | low N | low N |
| C: Wrap around PSM (L) | -0.19 | +0.14 | +0.01 | -0.17 | +0.16 | -0.05 | -0.01 | +0.21 | +0.08 |
| C: Wrap around PSM (R) | +0.09 | +0.11 | +0.16 | +0.06 | +0.26 | -0.51 | -0.02 | +0.29 | -0.46 |
| C: Pull/cinch (L) | -0.32 | -0.10 | -0.43 | -0.06 | -0.03 | +0.48 | -0.12 | -0.07 | +0.47 |
| C: Pull/cinch (R) | -0.35 | -0.37 | +0.16 | -0.21 | -0.18 | -0.08 | -0.20 | -0.18 | +0.08 |

Rank association: 1.00, 0.75, 0.50, 0.25, 0.00, −0.25, −0.50, −0.75, −1.00

Same layout and color scale in score_corr and score_corr_max. Original inference is preserved.
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